Next Toppers

ACP

Aarambh Chapterwise Problems

Class-10th

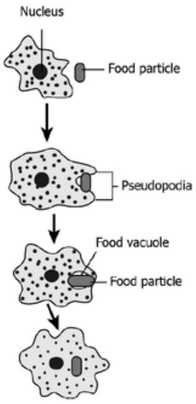
Life Processes

life processes

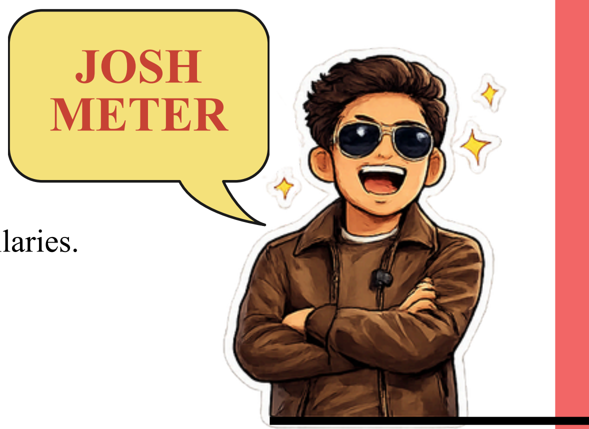
Read the question properly!

PART 1 (Basic MCQs)

1. The given image shows how Amoeba obtains nutrition.

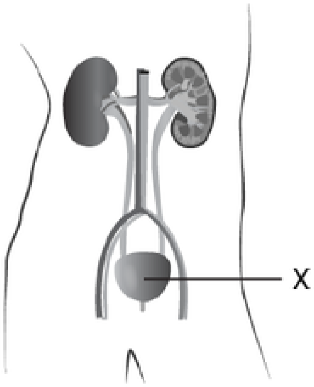


How is this process advantageous for amoeba?
(a) Capturing food takes less time
(b) Complex food can be digested easily
(c) More amount of food can be consumed
(d) Fast distribution of nutrition within the body



2. The alveoli in human lungs are surrounded by a network of blood capillaries. What is the main advantage of this arrangement?
(a) It helps in filtering dust particles from air
(b) It increases the rate of exchange of gases between air and blood
(c) It prevents the lungs from collapsing during breathing
(d) It helps in producing oxygen inside the lungs

3. The image shows the excretory system in humans.



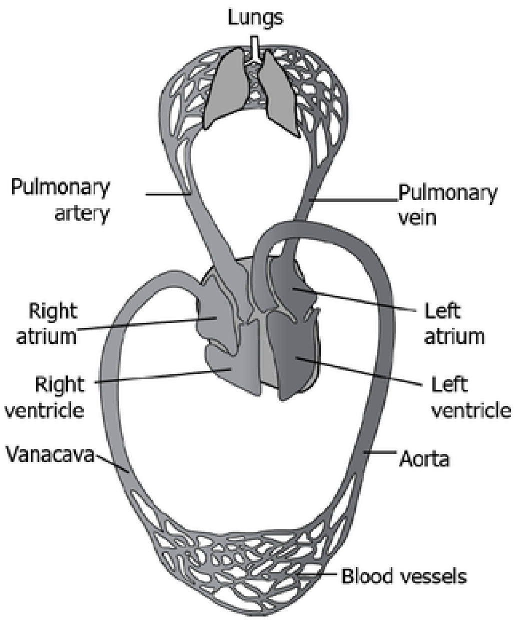
What is the importance of the labelled part in the excretory system?
(a) It produces urine.
(b) It filters waste from the blood.
(c) It stores the urine till urination.
(d) It carries urine from the kidney to the outside.

4. How is food transported from the phloem to the tissues according to plants' needs?
(a) Food is transported along with the water in the plant's body
(b) Food is transported in only one direction, like water in the plant body through the xylem
(c) Food is transported from a region with a low concentration to a higher concentration
(d) Food is transported from the region where it is produced to other parts of the plants

5.Which of the following is the correct sequence of body parts in the human alimentary canal?
(a) Mouth → stomach → small intestine → large intestine → oesophagus
(b) Mouth → oesophagus → stomach → small intestine → large intestine
(c) Mouth → stomach → oesophagus → small intestine → large intestine
(d) Mouth → oesophagus → stomach → large intestine → small intestine

6. Which of the following correctly explains why arteries have thick elastic walls?
(a) To store blood for long periods
(b) To withstand high pressure of blood pumped by the heart
(c) To allow exchange of gases
(d) To prevent clotting of blood

7. The image shows the transport of gases in the body through the heart and lungs.

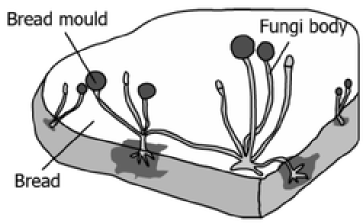


Which of the following option shows the transport of oxygen to the cell correctly?
(a) Lungs →pulmonary vein →left atrium →left ventricle →aorta → body cells
(b) Lungs →pulmonary vein →right atrium →right ventricle → aorta → body cells
(c) Lungs →pulmonary artery →left atrium → left ventricle → vena cava → body cells
(d) Lungs →pulmonary artery →right atrium → right ventricle→ vena cava → body cells

8. A plant gets rid of excess water through transpiration. What is the method used by plants to get rid of solid waste products?
(a) Shortening of stem
(b) Dropping down fruits
(c) Shedding of yellow leaves
(d) Expansion of roots into the soil

9. Which of the following are observed during anaerobic respiration in human muscle cells?
i) Absence of oxygen
ii) Release of energy
iii) Formation of lactic acid
iv) Formation of water
(a) i) and ii) only
(b) ii) and iii) only
(c) i), ii) and iii) only
(d) i), iii) and iv) only

10. The following image shows the bread moulds on bread:



How do these fungi obtain nutrition?
(a) By eating the bread on which it is growing
(b) By using nutrients from the bread to prepare their own food
(c) By breaking down the nutrients of bread and then absorbing them
(d) By allowing other organisms to grow on the bread and then consuming them

11. When a few drops of iodine solution are added to rice water, the solution turns blue-black in colour. This indicates that rice water contains:
(a) Fats
(b) Complex proteins
(c) Starch
(d) Simple proteins

12. In which of the following groups of organisms are food materials broken down outside the body and absorbed?
(a) Mushroom, green plants, amoeba
(b) Yeast, mushroom, bread mould
(c) Paramecium, amoeba, cuscuta
(d) Cuscuta, lice, tapeworm

13. Single circulation, i.e., blood flows through the heart only once during one cycle of passage through the body, is exhibited by which of the following:
(a) hyla, rana, draco
(b) whale, dolphin, turtle
(c) labeo, chameleon, salamander
(d) hippocampus, exocoetus, anabas

14. Name the pores in a leaf through which respi-ratory exchange of gases takes place.
(a) Lenticels
(b) Vacuoles
(c) Xylem
(d) Stomata

ASSERTION AND REASON TYPE QUESTION (1 Marks)

(yaha Marks katate h)

In the following questions a statement of Assertion is followed by a statement of Reason.
Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).
Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:
(a) Both A and R are true, and R is correct explanation of the assertion.
(b) Both A and R are true, but R is not the correct explanation of the assertion.
(c) A is true, but R is false.
(d) A is false, but R is true.

15. Assertion (A) : In plants there is no need of specialised respiratory organs.
Reason (R) : Plants do not have great demands of gaseous exchange.

16. Assertion (A) : Carbohydrate digestion mainly takes place in small intestine.
Reason (R) : Pancreatic juice contains the enzyme lactase.

17. Assertion : The movement of water and dissolved salts in xylem is always upwards.
Reason: 'The upward movement of water is due to low pressure created by transpiration.

18. Assertion : The Bowman's capsule and the tubule together make a nephron.
Reason : The function of tubule is to allow the selective reabsorption of substances like glucose, amino acids, urea, salts and water into the blood capillaries.

19. Assertion (A): In human beings, the respiratory pigment is haemoglobin
Reason (R) : It is a type of protein which has high-affinity carbon dioxide.

20. Assertion (A): The purpose of making urine is to filter out undigested food from intestine
Reason (R): Kidneys filter the waste and produce urine.

(ho jaayenge aaram se)

PART 2 (Subjective Questions)

SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

21. In a single-celled organism, diffusion is sufficient to meet all the requirements of food, gas exchange and waste removal, but it is not the case in multicellular organisms. Explain the reason for this difference.

22. How can haemodialysis save and prolong the life of uremic patients?

23. What are the three types of blood vessels? Mention one important feature of each.

24. A major portion of the carbohydrates produced by plants is stored in different parts of the plant (storage organs). Explain the mechanism by which this stored food is made available when different organs need it for growth.

25. Human beings exhibit 'double circulation' during which blood is passed through the lungs and heart.
(a) State the route of the first and the second circulation through the chambers of the heart and explain the usefulness of such circulation in humans.

(b) Name the blood vessels that:
(i) carry oxygenated blood from the lungs to the heart.
(ii) carry deoxygenated blood from the heart to the lungs.

26. A plant is kept in a humid environment where transpiration rate is very low. Despite sufficient water in the soil, the plant shows reduced upward movement of water.
(a) Identify the process responsible for upward movement of water in plants.
(b) Explain why this process is affected in humid conditions.
(c) Which tissue is responsible for this transport?
(d) Suggest one condition that would increase the efficiency of this process.

27. A plant kept in sunlight stops showing growth after its phloem is removed, even though photosynthesis continues. Explain why this happens.

28. During intense exercise, a person's breathing rate increases but oxygen supply still becomes insufficient. What type of respiration occurs in muscles and why?

29. If the kidneys fail to filter blood but the heart continues pumping normally, which life process is directly affected and how will it impact the body?

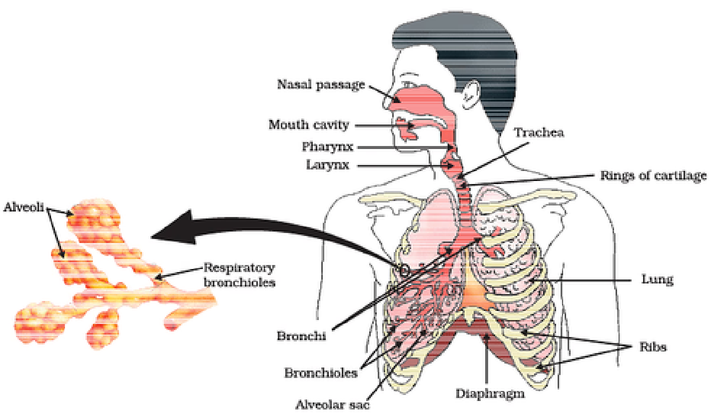
30. A person has normal digestion but still feels weak and loses weight continuously. Which step of nutrition is likely affected and why?

LONG ANSWER TYPE QUESTIONS (5 Marks)

(pahle points socho firr likho)

31. Aerobic respiration requires intake of oxygen to breakdown food to release energy.
(a) Name the structures through which gaseous exchange takes place in plants and human beings.
(b) Name the structure that controls the size of the chest cavity in humans to facilitate exchange of gases.
(c) What is the process by which gas exchange occurs in plants?
(d) Why is the process named in (c) not sufficient to carry oxygen throughout human body? How is this complemented in humans to ensure that oxygen is carried to all parts of the body?
(e) Reactions in living systems can absorb heat or release heat. State whether the heat energy is absorbed/ released during digestion. Also write the scientific term to denote the same.

32. Answer the following questions based on the diagram given below:



(a) Name the primary organs involved in the human respiratory system.
(b) What is the main function of the respiratory system?
(c) How does the diaphragm help in the process of breathing?
(d) What are alveoli, and why are they important in the respiratory system?
(e) Explain the role of cilia and mucus in the respiratory system.

33. Not all plants carry out photosynthesis by the same mechanism. In most plants, photosynthesis depends directly on the gaseous carbon dioxide that diffuses into the leaf through the stomata.
However, some plants — such as pineapple — have the ability to store carbon dioxide in the vacuoles of the leaf cells as part of a complex carbon compound. This complex compound is transported to the chloroplasts and releases carbon dioxide when required, for photosynthesis to occur.
This special photosynthesis mechanism is believed to have evolved as an adaptation to conserve water for survival in dry conditions.
(a) Which process in the plants does this photosynthesis mechanism minimise to help the plant survive in dry conditions?
(b) How is the ability to store carbon dioxide as a complex compound likely to help minimise the process referred to in (a)?
(c) When are such plants likely to take in carbon dioxide from the environment?

34. A student observes that a plant kept in sunlight grows well, but when the same plant is kept in a closed room without proper air circulation, its growth becomes weak even though it is watered regularly.
(a) Which life process is mainly affected in this situation?
(b) Explain why lack of proper air circulation affects this process.
(c) Name the structures involved in this process in plants.
(d) How does this condition indirectly affect another life process in plants?
(e) Suggest one method to improve the plant's growth in such conditions.

35. Describe the structure and functioning of nephrons.

36. Describe double circulation in human beings. Why is it necessary?

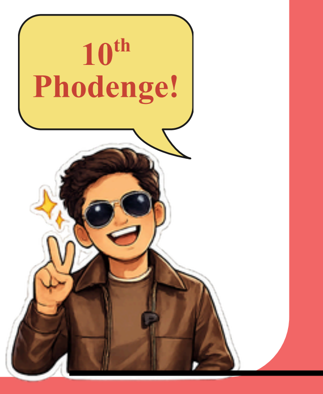
37. We often hear people complain about 'acidity' in the stomach.
(a) Overproduction of what substance is most likely the reason for the complaint?
(b) Why is the production of this substance necessary?
(c) How does the stomach prevent itself from the harmful effects of overproduction of the substance?

38. There are various muscles present in the human digestive system known as sphincters. Two examples of those are given below:
1. Pyloric sphincter – at the junction of stomach and small intestine
2. Anal sphincter – at the anus

Give ONE most likely consequence of malfunctioning of each of these sphincters.

39. A patient is suffering from kidney failure and is advised to undergo dialysis.
(a) Explain why dialysis is required in this condition.
(b) Describe the process of urine formation in a normal kidney.
(c) How is dialysis similar to and different from the function of a nephron?
(d) Name the main nitrogenous waste removed from the body..

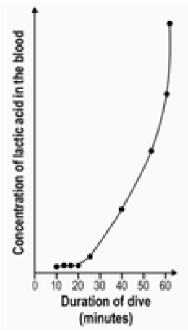
40. Arthropods and molluscs have a copper-containing respiratory pigment called haemocyanin while human beings have iron-containing haemoglobin.
(a) How do respiratory pigments help in the process of respiration?
(b) Why do multicellular animals need respiratory pigments?



PART 3 (Advanced Questions)

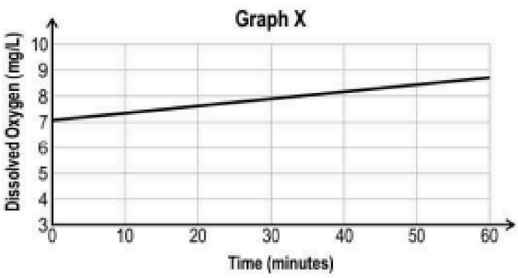
Case Study/Source Based Question (5 Marks)

41. A Weddell seal, a deep-sea diving mammal, has special adaptations that enable it to respire for long periods under water without inhaling air. Three such adaptations are given below:
P) When diving, the blood flow to all parts of the seal's body is reduced by 80–95%, except for a closed circuit between the lungs, heart and brain.
Q) 70% of the oxygen in the seal's body is stored in the blood (in haemoglobin), as compared to just 51% in humans.
R) 25% of the oxygen in the seal's body is stored in the muscles (in myoglobin), as compared to just 13% in humans.
In 1980, a group of scientists carried out an experiment to understand how a Weddell seal respires under water during dives of different durations. After each dive completed by the seal, they measured the concentration of lactic acid in the seal's blood. The graph below represents the data obtained by the scientists.

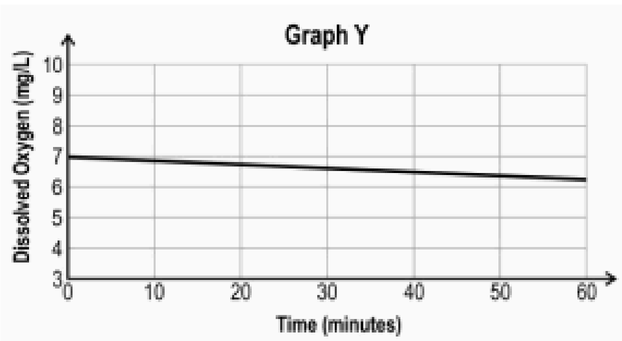


(a) Based on the graph, what can be inferred about the change in respiration happening in the seal's body after 20 minutes under water? Justify your answer.
(b) Adaptation R helps the seal the most to produce energy for swimming during the first 20 minutes of a dive. Explain why adaptations P and Q do not help as much.

42.Anita conducted an experiment to examine photosynthesis in aquatic plants kept in a tank by measuring dissolved oxygen. She plotted her results in the following graph X.



She repeated the experiment while covering the tank with an opaque black cloth. She plotted the results in the following graph Y.

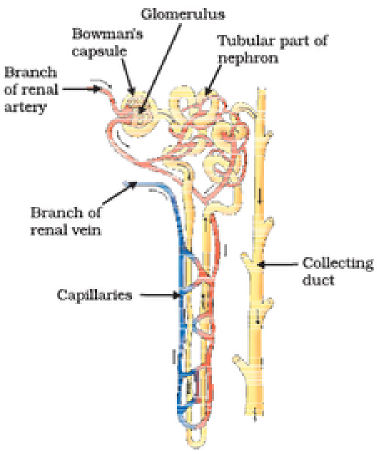


(a) What could be the aim of her experiment?
(b) Apart from photosynthesis, what other cellular process can be observed by the experiment?
(c) Why does the oxygen level rise in graph X?
(d) Explain the downward slope of the graph Y.

42. The image below shows the cross section of a blood vessel of a human arm.

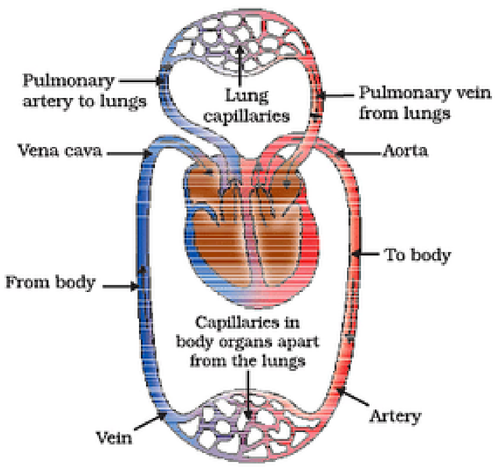
(a) What is the type of blood vessel shown in the image?
(b) Which direction will the blood flow in such a blood vessel?

43. Answer the following questions based on the diagram given below:



(a) What is the main function of a nephron in the human body?
(b) Name the two main parts of a nephron and briefly describe their functions.
(c) Which specific structure in the nephron is responsible for the initial filtration of blood?
(d) What is the role of the loop of Henle in the nephron?

44. Answer the following questions based on the diagram given below:

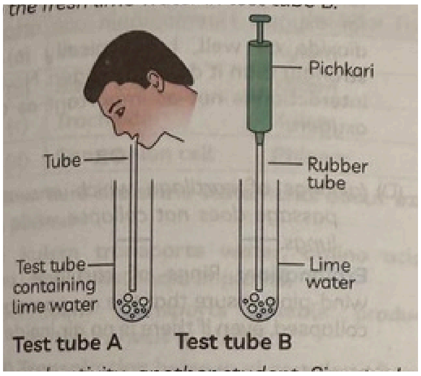


(a) What are the main organs involved in the exchange of oxygen and carbon dioxide as shown in the diagram?
(b) In the diagram, what represents the red blood cells? What is their role in this process?
(c) What does the blue arrow in the diagram indicate?
(d) Explain the role of the small blood vessels (capillaries) in the exchange of gases shown in the diagram.
(e) What is the significance of the arrows going in opposite directions between the lungs and body tissues in the diagram?

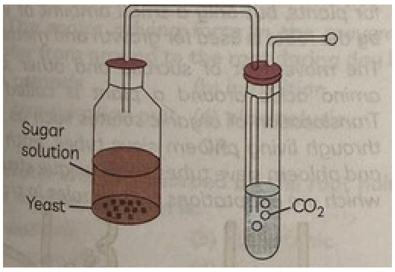
45. Divya went to see an aquarium along with her parents. She was surprised to see that the fish in the aquarium continuously opens and closes its mouth.

(A) Why does the fish do so?
(B) Divya holds her breath after breathing out for about 30–40 seconds. Is there any exchange of respiratory gases taking place in her lungs during this period? Explain.
(C) Write functions of the following parts:
(i) Rings of cartilage
(ii) Pharynx

46. In the first activity, a student, Rudra, took some freshly prepared lime water in two test tubes marked A and B. He blew air through the lime water in test tube A. He then used a syringe or pichkari and passed air through the fresh lime water in test tube B.

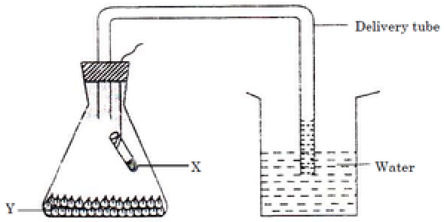


In the second activity, another student, Siya, took some fruit juice or sugar solution and added some yeast to this. She took this mixture in a test tube fitted with a one-holed cork and fitted the cork with a bent glass tube. She dipped the free end of the glass tube into a test tube containing freshly prepared lime water.



(A) In the first activity, what were the observations made by Rudra? What can be interpreted from his observations?
(B) Siya performed the second activity by taking yeast powder and fruit juice, she observed that lime water turned milky after few hours.
(i) What does this activity suggest to you?
(ii) What are the products formed during this activity?
(C) Name the energy currency in living organisms. When is it produced? Write one application of this energy currency produced in the cell.

47. In the experimental set-up shown:
(A) Name the material X filled in the small test tube and the material Y placed at the bottom of the conical flask.
(B) Why is there a rise in water level in the delivery tube?



- Nutrition – Q1, Q5, Q10, Q11, Q12, Q22, Q30
- Respiration – Q2, Q9, Q14, Q26, Q28, Q31, Q34, Q41, Q45, Q46, Q47
- Transportation in Humans – Q7, Q13, Q23, Q25, Q36, Q40, Q44
- Transportation in Plants – Q4, Q17, Q24, Q27, Q26
- Excretion – Q3, Q20, Q29, Q35, Q43
- Photosynthesis – Q8, Q28, Q33, Q42
- Stomata & Transpiration – Q8, Q24, Q33
- Transportation in Plants (Xylem & Phloem) – Q15, Q16, Q17, Q18, Q19, Q32
- Blood & Blood Vessels – Q6, Q23, Q42
- Double Circulation – Q7, Q25, Q36, Q41
- Digestion - Q37

SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

1. (b)
2. (b)
3. (c)
4. (d)
5. (b)
6. (b)
7. (a)
8. (c)
9. (c)

10 (c)
11 (c)
12 (b)
13 (d)
14 (d)
15 (a)
16 (c)
17 (a)

18 (c)
19. (c)
20. (d)

SHORT ANSWER TYPE QUESTIONS:

21. In single-celled organisms, diffusion is sufficient because all parts of the cell are in direct contact with the environment and distances are very small. In multicellular organisms, cells are far from the surface, so diffusion alone is not efficient. Therefore, specialised transport systems are required.

22. Haemodialysis helps in removing waste products like urea, excess salts, and water from the blood when kidneys fail. It maintains the chemical balance of the body and thus helps in prolonging the life of uremic patients.

23. The three types of blood vessels are:
- Arteries: Carry blood away from the heart; have thick muscular walls.
- Veins: Carry blood towards the heart; have valves to prevent backflow.
- Capillaries: Very thin walls; allow exchange of substances between blood and tissues.

24. Plants store carbohydrates mainly as starch. When required, starch is converted into soluble sugars and transported through phloem from storage organs (like roots or stems) to growing parts of the plant.

25. (a) First circulation: Right ventricle → lungs → left atrium
Second circulation: Left ventricle → body → right atrium
(b) Double circulation ensures efficient oxygen supply and prevents mixing of oxygenated and deoxygenated blood.

26. (a) Transpiration pull.
(b) In humid conditions, transpiration decreases because the air already has high water vapour, reducing the transpiration pull.
(c) Xylem.
(d) Low humidity or higher temperature increases transpiration and improves water transport.

27. Phloem transports food prepared in leaves to other parts of the plant. If phloem is removed, the food made during photosynthesis cannot reach growing regions like roots, buds and stem tips. Hence, growth stops even though photosynthesis continues.

28. During intense exercise, oxygen supply to muscles becomes insufficient. So muscle cells carry out anaerobic respiration. Glucose breaks down without oxygen to release a small amount of energy and forms lactic acid, which may cause cramps

29. The life process directly affected is excretion. If kidneys fail to filter blood, wastes like urea, excess salts and water accumulate in the body. This disturbs the body's internal balance and can become life-threatening.

30. The step likely affected is absorption or assimilation. If digested food is not properly absorbed in the small intestine, nutrients will not enter the blood. If assimilation is affected, nutrients will not be properly used by body cells, causing weakness and weight loss.

LONG ANSWER TYPE QUESTIONS:

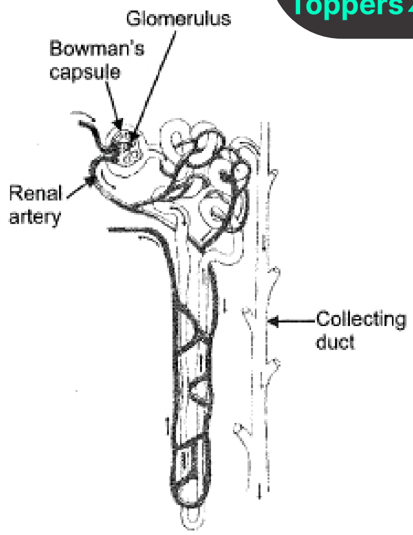
31. (a) Gas exchange occurs through stomata in plants and alveoli in human beings.
(b) The diaphragm controls the size of the chest cavity in humans.
(c) The process is diffusion.
(d) Diffusion alone is not sufficient in humans because body cells are far from the surface. It is complemented by the circulatory system, which transports oxygen to all parts of the body.
(e) During digestion, heat is released. Such reactions are called exothermic reactions.

32. (a) The primary organs involved in the human respiratory system are the lungs.
(b) The main function of the respiratory system is to facilitate the exchange of O_2 and CO_2.
(c) The diaphragm contracts and flattens during inhalation, increasing chest cavity volume and drawing air into the lungs; it relaxes during exhalation, pushing air out.
(d) Alveoli are tiny air sacs in the lungs where gas exchange occurs; their thin walls and large surface area make exchange efficient.
(e) Cilia move mucus and trapped particles out of the respiratory tract, while mucus traps dust, bacteria, and other foreign particles.

33. (a) This mechanism minimises transpiration (water loss).
(b) Storing CO_2 allows stomata to remain closed during the day, reducing water loss.
(c) Such plants take in CO_2 during the night.

34. (a) The life process mainly affected is photosynthesis.
(b) Proper air circulation provides carbon dioxide, which is essential for photosynthesis. In a closed room, carbon dioxide becomes limited, reducing the rate of photosynthesis.
(c) The structures involved are stomata, which help in gaseous exchange.
(d) Reduced photosynthesis leads to less glucose formation. This decreases respiration, resulting in less energy and weak plant growth.
(e) The plant should be kept in a place with proper ventilation or fresh air to ensure sufficient carbon dioxide supply.

35. Structure of nephrons: It consists of a Bowman's capsule in which glomerulus is present (cluster of capillaries). The afferent artery brings the impure blood to nephron. The cup shaped structure (Bowman's capsule) form a tubular part of nephron which leads to collecting duct.



Structure of a nephron

(i) Filtration: The renal artery or afferent artery is wider and slowly it becomes a narrow tube in the
glomerulus. Due to difference in the width, pressure difference is caused and water with dissolved impurities are squeezed out from the tube. It is collected in the Bowman's capsule which is cup like structure and passes into the tube.
(ii) Reabsorbtion: The above filtrate passes through the tubule where the major amount of water, glucose, amino acids are selectively reabsorbed by the capillaries which are surrounding the tubule.
(iii) Urine formation: The water and impurities which is not reabsorbed is sent to a collecting duct. This filtrate contains more of dissolved nitrogenous wastes i.e. urea and hence it is termed as urine. From here the urine enters the ureter and is collected in urinary bladder.

36. The heart of human beings consist of two sides right and left. The right side of the heart receives de-oxygenated blood from the cells and tissues and sends it further for purification to lungs. The left side of the heart receives oxygenated blood from lungs which is pumped further and sent to all the parts of the body through blood vessels. This is called double circulation. The energy demand of human beings is too large and hence it is necessary for the separation of oxygenated and deoxygenated blood to meet this energy demand.

37. (a) Caused by excess hydrochloric acid (HCl)
(b) HCl creates an acidic medium which helps pepsin digest proteins and kills harmful bacteria.
(c) The stomach protects itself by secreting mucus, which prevents damage from acid.

38.
- Pyloric sphincter: Improper movement of food into small intestine → digestion problems
- Anal sphincter: Loss of control over defecation

39. (a) Why dialysis is required: Dialysis is required when the kidneys fail to function properly. In kidney failure, harmful waste products like urea, excess salts, and water accumulate in the blood. Dialysis acts as an artificial kidney and helps in removing these toxic substances, thus maintaining the chemical balance of the body.
(b) Process of urine formation in a normal kidney: Urine formation takes place in nephrons in three steps:
1. Ultrafiltration: Blood is filtered in the glomerulus; small molecules like urea, salts, and water pass into the tubule.
2. Reabsorption: Useful substances like glucose, amino acids, and some water are reabsorbed into the blood.
3. Secretion: Additional wastes and excess ions are secreted into the tubule.
4. The remaining fluid forms urine.

(c) Similarity and difference between dialysis and nephron:
Similarity:
- Both remove waste products like urea from the blood.
- Both help in maintaining water and salt balance.

Difference:
- Nephron is a natural filtering unit of the kidney, while dialysis is an artificial process.
- Nephron performs selective reabsorption and secretion, while dialysis mainly filters wastes using a machine and semi-permeable membrane.

(d) Main nitrogenous waste: The main nitrogenous waste removed from the body is urea.

40. (a) Respiratory pigments (like haemoglobin) bind oxygen and help transport it in the body.
(b) Multicellular organisms need them because diffusion alone is not sufficient to supply oxygen to all body cells.

CASE STUDY/SOURCE BASED QUESTION:

41. (a) From the graph, lactic acid concentration increases sharply after about 20 minutes.
This indicates that the seal shifts from aerobic respiration to anaerobic respiration due to lack of oxygen.
(b) During the first 20 minutes, the seal mainly uses oxygen stored in:

- Blood (haemoglobin)
- Muscles (myoglobin)

Adaptations Q and R help store oxygen, so energy is produced aerobically initially.
Hence, adaptations P (reduced blood flow) is less important in early stages.

COMPETENCY BASED QUESTION:

42. Experiment on Photosynthesis (Graphs X & Y)
(a) Aim: To study the effect of light on photosynthesis by measuring dissolved oxygen.
(b) Apart from photosynthesis, respiration is also observed.
(c) In graph X, oxygen level increases because photosynthesis is occurring in presence of light, releasing oxygen.
(d) In graph Y, oxygen level decreases because the tank is covered (no light), so photosynthesis stops and only respiration occurs, which consumes oxygen.

43. (a) Main function: Filtration of blood and formation of urine
(b) Two main parts:

- Glomerulus: Filters blood (ultrafiltration)
- Tubule (renal tubule): Reabsorption of useful substances and secretion of wastes

(c) Initial filtration occurs in the glomerulus
(d) Loop of Henle helps in reabsorption of water and salts, concentrating urine.

44. (a) Main organs: Heart and lungs
(b) Red blood cells contain haemoglobin, which carries oxygen from lungs to body tissues.
(c) Blue arrows indicate deoxygenated blood flowing from body to lungs.
(d) Capillaries are thin-walled vessels that allow exchange of oxygen and carbon dioxide between blood and tissues.
(e) Opposite arrows show double circulation:

- Oxygenated blood goes from lungs to body
- Deoxygenated blood returns from body to lungs
- This ensures efficient oxygen supply and removal of carbon dioxide.

45. (A) Fish breathe through gills. They open and close their mouth to take in water, which passes over the gills where oxygen is absorbed and carbon dioxide is released.
(B) Yes, gas exchange continues. Some air remains in the lungs (residual volume), which allows exchange of oxygen and carbon dioxide for a short time.
(C) (i) Rings of cartilage: Prevent the air passage from collapsing.
(ii) Pharynx: Acts as a common passage for air and food.

46. (A) Observation: Lime water in test tube A turns milky faster than in test tube B.
Interpretation: Exhaled air contains more carbon dioxide than atmospheric air.

(B) (i) It shows that yeast carries out respiration (fermentation) and releases carbon dioxide.
(ii) Products formed: Carbon dioxide and alcohol (ethanol).
(C) Energy currency: ATP (Adenosine Triphosphate).
Produced during cellular respiration.
Application: Used for activities like muscle movement.

47. (A) X: KOH pellets
Y: Wet germinating seeds
(B) Germinating seeds use oxygen present in the flask during respiration and release carbon dioxide. The carbon dioxide released is absorbed by KOH pellets. Due to this, a partial vacuum is created inside the flask, so water rises in the delivery tube.